

WhatsApp Investigation Intelligence Platform (WIIP)

Executive Summary

Digital forensic tools are highly effective at extracting evidence from mobile devices. However, investigators still spend significant time manually correlating conversations, reconstructing timelines, identifying relationships, and understanding the context behind an incident.

Vision

Become the world's leading investigation intelligence platform capable of transforming digital evidence into actionable investigation insights.

Mission

- Reduce investigation time.
- Eliminate manual evidence correlation.
- Reconstruct digital communication history.
- Reveal hidden relationships.
- Detect communication behavior changes.
- Support evidence-driven investigations.
- Maintain explainable and court-defensible analysis.

Product Philosophy

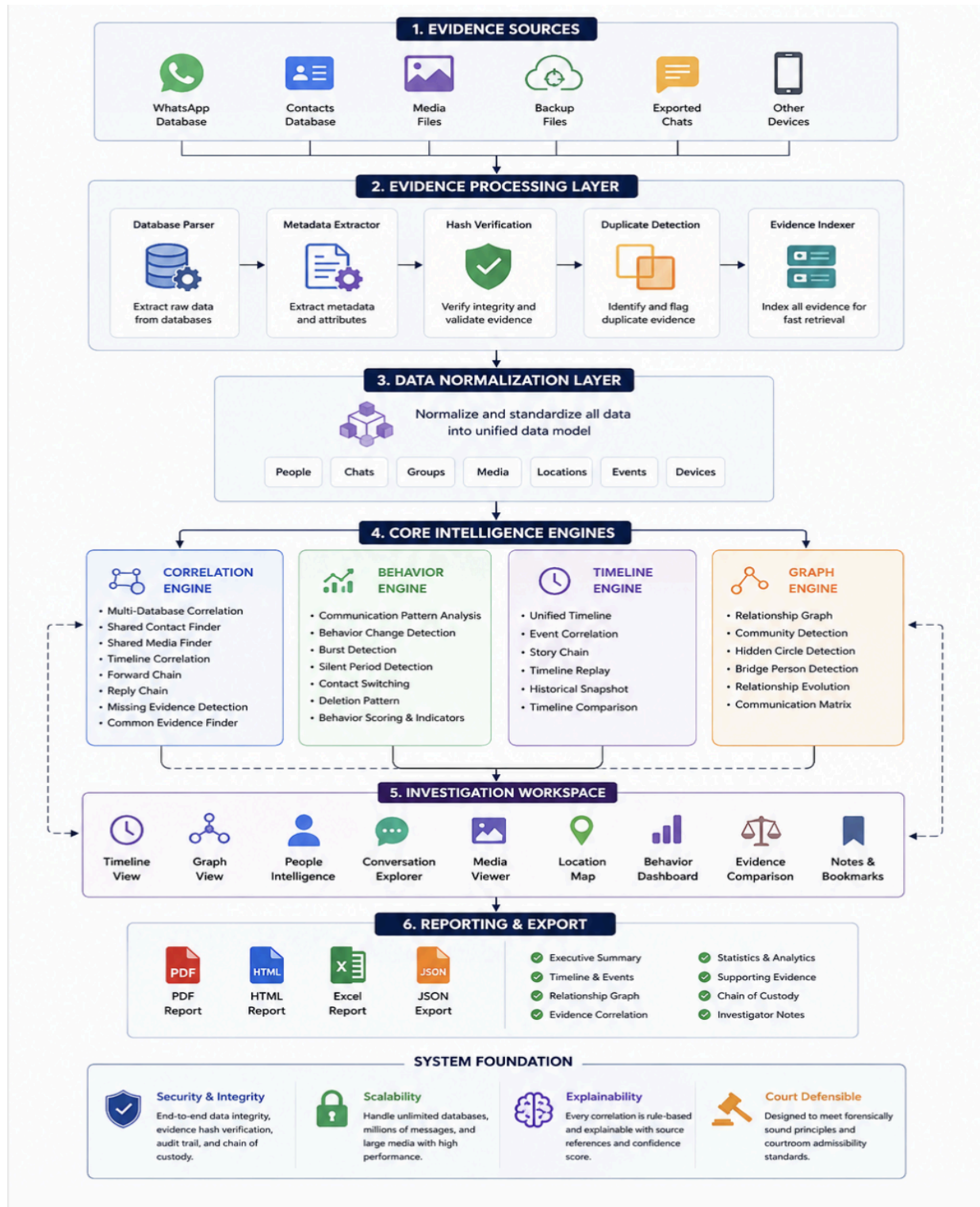
Traditional forensic software answers:

"What evidence exists?"

WIIP answers:

"How are all pieces of evidence connected?"

Core Architecture



System Foundation

WIIP is built on four core intelligence engines.

Every feature inside the platform is generated from these engines, allowing the system to remain scalable, explainable, and extensible.

Correlation Engine

Purpose

Connect every digital artifact across unlimited WhatsApp databases.

Supported Objects

- Person
- Phone Number
- Device
- Chat
- Group
- Media
- Voice Note
- Call
- Location
- Document
- Hyperlink
- Timestamp

Capabilities

- Multi-Database Correlation
- Shared Contact Finder
- Shared Group Finder
- Shared Media Finder
- Shared Voice Note Finder
- Shared Document Finder
- Shared Location Finder
- Shared Link Finder
- Duplicate Evidence Detection
- Timeline Correlation
- Message Comparison

- Missing Evidence Detection
- Missing Message Detection
- Forward Chain
- Reply Chain
- Quote Chain
- Cross-Database Comparison

Behavior Engine

Purpose

Analyze digital communication behavior rather than only message content.

Behavior itself becomes digital evidence.

Communication Pattern Analysis

- Daily Activity Pattern
- Weekly Activity Pattern
- Monthly Activity Pattern
- Active Hour Distribution
- Peak Communication Hours
- Communication Frequency
- Response Time Pattern
- Conversation Duration
- Call Frequency
- Media Sharing Frequency

Behavior Change Detection

- Communication Burst Detection
- Silent Period Detection
- Contact Appearance
- Contact Disappearance
- Contact Switching
- Sudden Activity Increase
- Sudden Activity Drop
- Group Join Pattern
- Group Leave Pattern
- Message Deletion Pattern
- Forwarding Pattern
- Location Sharing Pattern

Behavior Indicators

- Activity Score
- Communication Stability
- Interaction Score
- Consistency Score
- Relationship Strength

Timeline Engine

Purpose

Convert isolated evidence into one continuous chronological investigation timeline.

Timeline Objects

- Messages
- Calls
- Images
- Videos
- Voice Notes
- Documents
- Shared Locations
- Deleted Messages
- Forwarded Messages
- Replies
- Quotes

Capabilities

- Unified Timeline
- Historical Snapshot
- Timeline Replay
- Timeline Comparison
- Event Correlation
- Story Chain
- Investigation Replay

Graph Engine

Purpose

Represent every evidence object as an interconnected investigation graph.

PERSON

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Message Call Group

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Media Location Device

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Timeline Event

Capabilities

- Relationship Graph
- Family Mapping
- Community Detection
- Hidden Circle Detection
- Contact Intersection
- Group Intersection
- Relationship Evolution
- Bridge Person Detection
- Communication Matrix

Product Modules

Case Management

Manage investigation lifecycle.

Features:

- Case Dashboard
- Evidence Manager
- Investigator Assignment
- Chain of Custody
- Audit Trail
- Evidence Verification
- Evidence Hash Verification
- Investigation Notes
- Bookmark Evidence
- Tagging System
- Case Status

Evidence Processing

Import and validate digital evidence.

Supported Sources:

- WhatsApp Database
- Contacts Database
- Media Files
- Backup Files
- Exported Chats

Processing Features:

- Database Parsing
- Metadata Extraction
- Evidence Indexing
- Duplicate Detection
- Hash Verification

People Intelligence

Build a complete profile for every phone number.

Features:

- Contact Profile
- Aliases
- Communication Count
- Active Days
- Active Hours
- First Seen
- Last Seen
- Favorite Contacts
- Favorite Groups
- Shared Evidence
- Shared Locations

Conversation Intelligence

Analyze communication content.

Features:

- Chat Explorer
- Topic Explorer
- Keyword Search
- Conversation Statistics
- Longest Conversation
- Most Active Conversation
- Most Active Day
- Communication Heatmap

Media Intelligence

Analyze multimedia evidence.

Features:

- Media Browser
- Duplicate Images
- Duplicate Videos
- Duplicate Documents
- Shared Media
- OCR
- Metadata Viewer
- Media Timeline

Location Intelligence

Visualize location evidence.

Features:

- Map View
- Location Timeline
- Meeting Point Detection
- Shared Location Analysis
- Common Locations
- Route History
- Geo Correlation

Investigation Workspace

A unified workspace for investigators.

Investigators can pin:

- Contacts
- Messages
- Calls
- Photos
- Videos
- Voice Notes
- Documents
- Locations
- Groups

Everything remains synchronized with the Timeline, Graph, and Behavior Engines.

Reporting

Generate investigation-ready reports.

Supported Formats:

- PDF
- HTML
- Excel
- JSON

Report Contents:

- Executive Summary
- Timeline

- Relationship Graph
- Evidence Summary
- Correlation Results
- Supporting Evidence
- Chain of Custody

Killer Features

Live Correlation Board

Every interaction immediately updates:

- Timeline
- Relationship Graph
- Evidence List
- Statistics
- Behavior Dashboard

No manual re-analysis is required.

Behavior Intelligence

Communication behavior is treated as evidence.

Investigators can immediately identify:

- Communication spikes
- Silent periods
- New contacts
- Disappearing contacts
- Message deletion trends
- Group activity changes
- Communication rhythm changes
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Story Chain

Instead of isolated messages, investigators see complete communication sequences.

Message

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Reply

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Call

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Shared Media

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Shared Location

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Deleted Message

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Communication Stops

Historical Snapshot

Reconstruct the investigation exactly as it appeared on any selected date.

Includes:

- Timeline
- Graph
- Contacts
- Groups
- Media
- Statistics

Case Replay

Replay the investigation chronologically.

08:10 Message Sent

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08:12 Reply Received

|

08:15 Voice Call

|

08:19 Photo Shared

|

08:24 Location Shared

|

08:30 Message Deleted

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08:35 Communication Ends

Relationship Evolution

Visualize how communication networks change over time.

Investigators can compare relationships:

- Before Incident
- During Incident
- After Incident

Hidden Circle Detection

Automatically detect hidden communication communities by analyzing shared contacts, shared groups, communication frequency, and interaction patterns.

Common Evidence Finder

Automatically identify identical evidence across unlimited databases.

Supported Evidence:

- Images
- Videos

- Voice Notes
- Documents
- Contacts
- Groups
- Locations

Multi-Database Timeline

Merge unlimited WhatsApp databases into one synchronized investigation timeline.

Investigation Gap Analyzer

Reveal missing evidence instead of only displaying existing evidence.

Examples:

- Missing witness device
- Timeline gaps
- Communication blackout
- Missing database
- Missing media
- Missing contacts

Cross-Case Correlation

Correlate investigation entities across previous cases.

Detect:

- Recurring phone numbers
- Recurring aliases
- Recurring media hashes
- Recurring locations
- Recurring groups
- Recurring communication patterns

Evidence Confidence Engine

Every correlation includes:

- Supporting Evidence Count
- Correlation Strength
- Source Database Reference

- Timestamp Consistency
- Direct Evidence Reference

This ensures every finding remains transparent and independently verifiable.

Product Workflow

Evidence Collection

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Evidence Validation

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Data Processing

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Correlation Analysis

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Behavior Analysis

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Timeline Reconstruction

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Relationship Mapping

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Investigation Workspace

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Evidence Review

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Court-Ready Report

Product Positioning

It is an Investigation Intelligence Platform that transforms fragmented digital evidence into a connected investigation ecosystem.

Every message, person, media file, location, timestamp, device, and communication event becomes part of a single investigation graph, allowing investigators to understand not only what happened, but how every piece of evidence is connected.

Tagline

Beyond WhatsApp Forensics. Built for Investigation Intelligence.